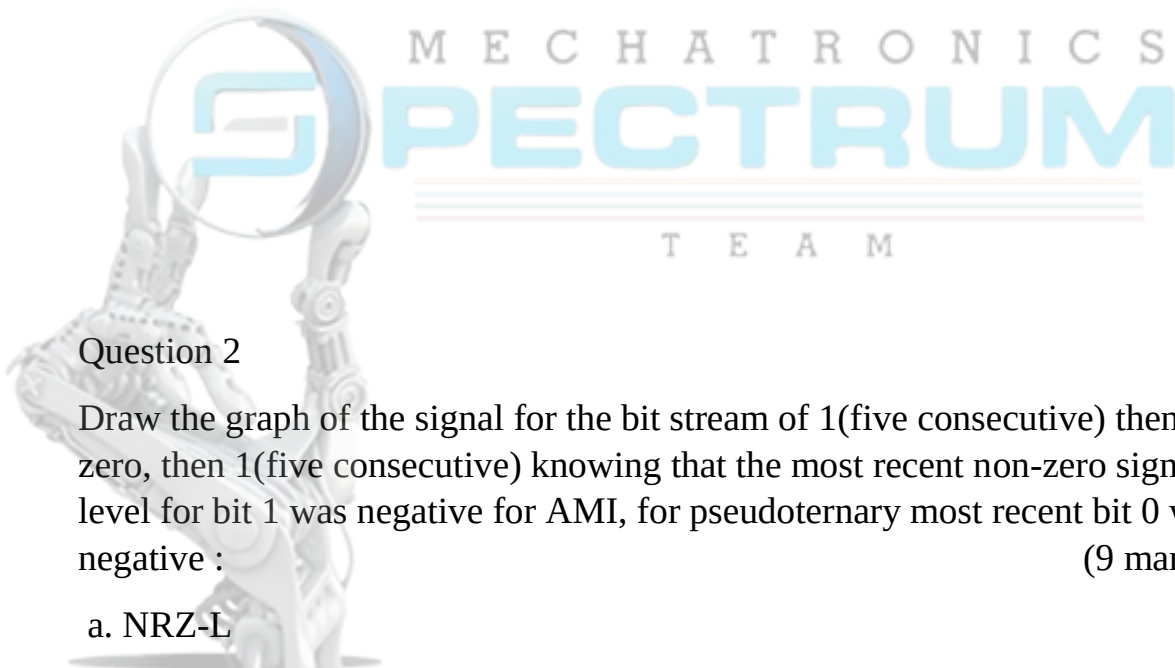


### Question 1

Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse-stuffing TDM with no synchronization bits. Answer the following questions: (12 marks)

- a. What is the size of the frame?
- b. What is the frame rate?
- c. What is the duration of the frame?
- d. What is the data rate of the link?



### Question 2

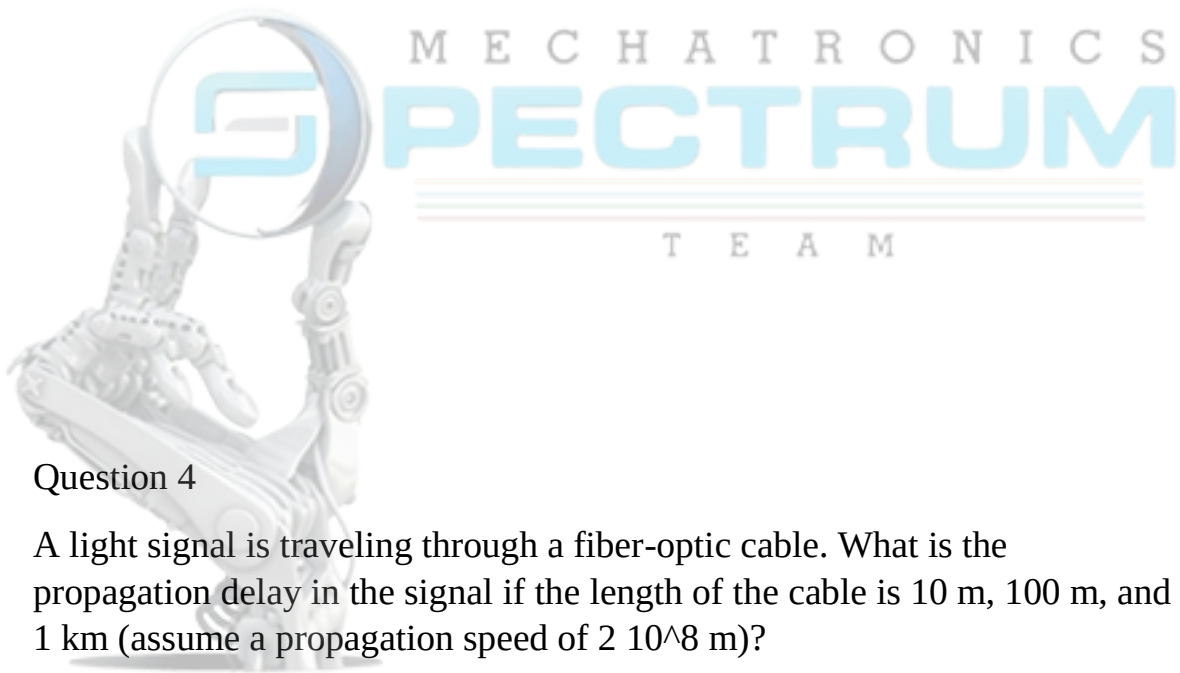
Draw the graph of the signal for the bit stream of 1(five consecutive) then a zero, then 1(five consecutive) knowing that the most recent non-zero signal level for bit 1 was negative for AMI, for pseudoternary most recent bit 0 was negative : (9 marks)

- a. NRZ-L
- b. Bipolar AMI
- c. Psuedoternary

### Question 3

A voice signal with a frequency range from 300 Hz to 3000 Hz is sampled at a rate of 7000 samples per second.

- a. If  $\text{SNR}=30\text{dB}$ , find the number of quantization levels ( $L$ ).
- b. What is the recommended data rate?



### Question 4

A light signal is traveling through a fiber-optic cable. What is the propagation delay in the signal if the length of the cable is 10 m, 100 m, and 1 km (assume a propagation speed of  $2 \cdot 10^8$  m)?

### Question 5

How many bits per baud can be sent in each of the following cases if the signal constellation has the given number of points? (11 marks)

- a. 2      b. 4      c. 16      d. 1024

